

HiveForce Labs

THREAT ADVISORY

 **ATTACK REPORT**

Serpents in Disguise: MuddyWater's Hidden Toolset Exposed

Date of Publication

December 3, 2025

Admiralty Code

A1

TA Number

TA2025366

Summary

Attack Discovered: September 30th, 2024

Targeted Countries: Israel and Egypt

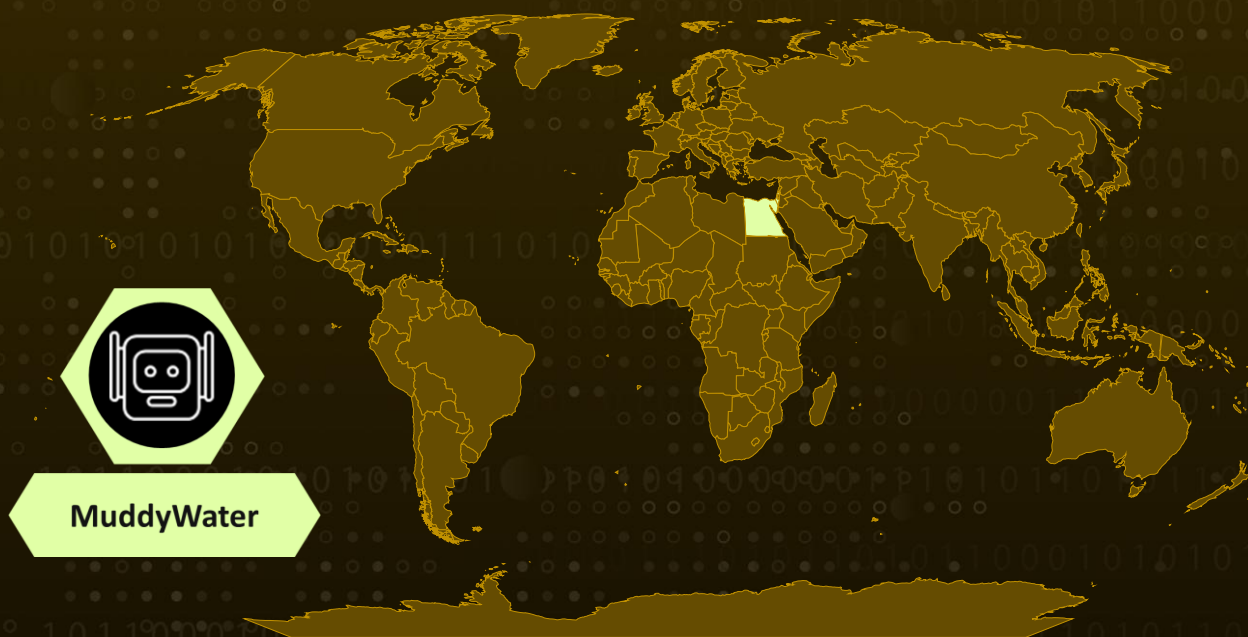
Targeted Industries: Technology, Engineering, Government, Manufacturing, Transportation, Utilities, University

Malware: Fooder loader, MuddyViper, CE-Notes, LP-Notes, Blub, go-socks5

Actor: MuddyWater (aka Seedworm, TEMP.Zagros, Static Kitten, Mercury, TA450, Cobalt Ulster, ATK 51, T-APT-14, ITG17, Mango Sandstorm, Boggy Serpens, Yellow Nix, G0069)

Attack: MuddyWater's latest campaign marks a noticeable shift from their historically noisy style to a far more careful and calculated espionage effort. Disguising malware as simple games and trusted tools, the group quietly infiltrated organizations in Israel and Egypt using spearphishing links, deceptive RMM installers, and a growing arsenal of custom loaders and backdoors. Their new tools, like the Snake-themed Fooder loader, the stealthy MuddyViper backdoor, and upgraded credential stealers, showcase a maturing playbook focused on persistence, credential harvesting, and covert communication. With refined techniques, smarter evasion, and a broader toolkit blending custom malware with publicly available components, MuddyWater is steadily evolving into a more disciplined and dangerous espionage actor.

🔪 Attack Regions



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Attack Details

#1

MuddyWater, a long-running Iran-aligned cyberespionage group, has resurfaced with a more polished and stealthy campaign targeting organizations in Israel and Egypt. Known historically for noisy operations and reliance on familiar toolsets, the group has now incorporated a suite of previously undocumented custom tools that significantly elevate their tradecraft. Central to this campaign is Fooder, a deceptive loader that poses as a simple Snake game while quietly executing MuddyViper, a sophisticated C/C++ backdoor. Together, these components enable attackers to harvest system information, execute commands, manipulate files, and steal sensitive credentials, all while maintaining a low profile.

#2

Active since 2017, MuddyWater has traditionally targeted government, telecommunications, energy, and critical infrastructure sectors across the Middle East and North America. Their recent activity campaigns show the group adopting multi-stage payloads, leveraging remote monitoring and management (RMM) tools for access, and deploying better-engineered malware, hinting at a more structured development pipeline and a deeper investment in operational security.

#3

The latest wave, spanning September 2024 to March 2025, relied heavily on spearphishing emails directing victims to file-sharing services such as OneHub, Egnyte, and Mega to download trojanized RMM installers. Once inside a network, MuddyWater employed predictable, yet effective backdoors written in PowerShell and Go, alongside more deceptive implants like the VAX-One backdoor masquerading as legitimate software. Despite their improved stealth, MuddyWater continues to reuse recognizable coding patterns and infrastructure, making attribution straightforward for researchers who have followed their activity over the years.

#4

It was uncovered compelling code overlaps linking new components such as LP-Notes, CE-Notes, MuddyViper, and various go-socks5 reverse tunnels to previously documented MuddyWater activity. These tools share encryption approaches, UI spoofing techniques, and even design quirks unique to Iranian-aligned groups. Fooder stands out as the most innovative, using reflective loading, AES-based payload decryption, and even game-like delay mechanics to mask malicious execution. Meanwhile, its payload, MuddyViper, reveals a detailed picture of compromise, offering attackers 20 distinct commands to manipulate victim systems, establish persistence, harvest browser credentials, deploy reverse shells, and impersonate Windows security prompts to steal user passwords.

#5

Taken together, this campaign reflects a clear maturation in MuddyWater's capabilities. While remnants of their traditionally noisy operations remain, the introduction of purpose-built loaders, cleverly disguised interfaces, and improved credential-theft mechanisms highlights a group working to refine its espionage toolkit. [MuddyWater](#) continues to expand its role within Iran-nexus cyber operations. Ongoing monitoring will be crucial, as the group's evolving toolset suggests further enhancements and broader targeting in the months ahead.

Recommendations



Strengthen Email Awareness and Verification: Since MuddyWater relies heavily on spearphishing links and fake RMM installers, make sure your teams know how to spot suspicious emails, especially those asking them to download software. A quick verification with IT before opening anything can stop the attack chain right at the start.



Prioritise Endpoint Monitoring for Unusual Loaders and Scripts: Tools like Fooder and MuddyViper run in memory and mimic normal applications. Enable EDR products and monitor for odd process launches, reflective loading, PowerShell scripts, or unknown executables masquerading as legitimate software.



Harden Browser Credential Storage: Because MuddyWater actively steals browser passwords, encourage users to migrate sensitive credentials to dedicated password managers and disable the storage of corporate passwords in browsers. Enforce MFA to reduce the impact of any stolen credentials.



Regularly Review Persistence Mechanisms: MuddyViper sets scheduled tasks or Startup entries to stay active. Periodically scan for unusual scheduled tasks, Startup folder additions, or unsigned binaries running at boot.



Enhance Endpoint Protection: Deploy next-generation antivirus (NGAV) and endpoint detection & response (EDR) solutions to identify and block malware. Leverage behavioral analysis and machine learning-based detection to spot suspicious activity.



Potential MITRE ATT&CK TTPs

<u>TA0043</u> Reconnaissance	<u>TA0042</u> Resource Development	<u>TA0001</u> Initial Access	<u>TA0002</u> Execution
<u>TA0003</u> Persistence	<u>TA0005</u> Defense Evasion	<u>TA0006</u> Credential Access	<u>TA0007</u> Discovery
<u>TA0009</u> Collection	<u>TA0010</u> Exfiltration	<u>TA0011</u> Command and Control	<u>T1591</u> Gather Victim Org Information

<u>T1583</u> Acquire Infrastructure	<u>T1608</u> Stage Capabilities	<u>T1587</u> Develop Capabilities	<u>T1587.001</u> Malware
<u>T1588</u> Obtain Capabilities	<u>T1588.002</u> Tool	<u>T1566</u> Phishing	<u>T1566.002</u> Spearphishing Link
<u>T1059</u> Command and Scripting Interpreter	<u>T1059.001</u> PowerShell	<u>T1059.003</u> Windows Command Shell	<u>T1559</u> Inter-Process Communication
<u>T1559.001</u> Component Object Model	<u>T1106</u> Native API	<u>T1204</u> User Execution	<u>T1204.001</u> Malicious Link
<u>T1547</u> Boot or Logon Autostart Execution	<u>T1547.001</u> Registry Run Keys / Startup Folder	<u>T1543</u> Create or Modify System Process	<u>T1543.003</u> Windows Service
<u>T1053</u> Scheduled Task/Job	<u>T1134</u> Access Token Manipulation	<u>T1134.001</u> Token Impersonation/Theft	<u>T1140</u> Deobfuscate/Decode Files or Information
<u>T1620</u> Reflective Code Loading	<u>T1497</u> Virtualization/Sandbox Evasion	<u>T1497.003</u> Time Based Checks	<u>T1027</u> Obfuscated Files or Information
<u>T1027.007</u> Dynamic API Resolution	<u>T1134.002</u> Create Process with Token	<u>T1622</u> Debugger Evasion	<u>T1070</u> Indicator Removal
<u>T1622</u> Clear Persistence	<u>T1070.004</u> File Deletion	<u>T1036</u> Masquerading	<u>T1036.004</u> Masquerade Task or Service
<u>T1112</u> Modify Registry	<u>T1027.009</u> Embedded Payloads	<u>T1027.013</u> Encrypted/Encoded File	<u>T1555</u> Credentials from Password Stores
<u>T1555.003</u> Credentials from Web Browsers	<u>T1056</u> Input Capture	<u>T1056.002</u> GUI Input Capture	<u>T1082</u> System Information Discovery
<u>T1518</u> Software Discovery	<u>T1518.001</u> Security Software Discovery	<u>T1074</u> Data Staged	<u>T1074.001</u> Local Data Staging

T1560 Archive Collected Data	T1560.001 Archive via Utility	T1573 Encrypted Channel	T1573.001 Symmetric Cryptography
T1219 Remote Access Tools	T1071 Application Layer Protocol	T1071.001 Web Protocols	T1105 Ingress Tool Transfer
T1001 Data Obfuscation	T1090 Proxy	T1041 Exfiltration Over C2 Channel	T1030 Data Transfer Size Limits

🔪 Indicators of Compromise (IOCs)

TYPE	VALUE
SHA1	76632910CF67697BF5D7285FAE38BFCF438EC082, 1723D5EA7185D2E339FA9529D245DAA5D5C9A932, 69B097D8A3205605506E6C1CC3C13B71091CB519, B7A8F09CB5FF8A33653988FFBA585118ACF24C13, B8997526E4781A6A1479690E30072F38E091899D, 8E21DE54638A79D8489C59D958B23FE22E90944A, CD47420F5CE408D95C98306D78B977CDA0400C8F, C1299E8C9A8567A9C292157F3ED65B818AA78900, 29CDA06701F9A9C0A6791775C3EB70F5B52BBEFF, 8F3ED626E7B929450E36E97BA5539C8371DF0EF8, 007B5CD6D6ACF972F7743F79E23CAB9BB2ECBEE3, CD36F93DBC4C718930593D8F029EFDCAA52B619B, 47B70C47BEB33E88B4197D6AF1B768230E51B067, D46900D78AE036967E0B37F9EC6A8000131AE604, 0657D0B0610618886DDD74C3D0A1D582CDD24863, 2939FD218E0145D730BD94AA1C76386A5259EACE, 3BC6502A55A4D5D29132DA4D9943E154A810CC83, 7950296331802188EB99E232E2C383CB9FDD5D7D, 8580824FE14DB158388102B16C1C79DFBBA36083, B48B93B4EB69D01588D371356EDE614C5E7378DE, EA8A1C2382FF765709D7F78EF60482598E4C0DEB, EAF4BAFC62170C9FCA1F6B591848883DBF97F93D, F5EFBA6CCBA5A6AD6C3AFA928C0E5EAA44597411, 13DA612D75DC5268F5235F5BACE6D8F0DB0091FF, 25361183DE63F296BA71B6FCF0725E022B3C989A, 0E9A4892CFA1C9065B36D8F2E164E28609A8CF5D, 2B09241CA025BDC4455E9F6BA6009E2F27C08EDF, 2E9BE23CDD8152DB6CD1A54E001C4EA82FF6F1C6, 45FA7DE711FEA1F8D1E348E87834246C455DD2ED, 4E0EF2386980639FC5355FD68DAFF54EB2AD622E,

TYPE	VALUE
SHA1	4E9529BA4A6E42D6278D37E3FDEE9E1D991CEBE0, 50C6D4A2AD16A231CF11C43F3BBC868D90E20D25, 52009F36058337B6401DA0A0F4885A0C185F0520, 535882B6EDAB29247E035236A84CA510FB1E0854, 544CE18E4C1F1B288DEE6018DFCF4E4D4A315F7A, 54EBC125039CC83E4682CA44DD592534562B25C3, 5A08150C1DC17E9F691296F0A577C2EC9BA8028C, 5D1E61DA8083C41FF1FC23A1222A4A88B43A4E9B, 6532E0437C8913FA418F1EE258561B15BBEE9052, 6CA41565844118385B345A39A9B79E0BBC0DD338, 6FC50A99AAE1D6C40111632D4F49BD19F9794CF6, 826CFF5D85713CE4B2F3C15AB53A84E6848D2E2C, 87ADD79C7C8335447113EE0D413F52AE2B17F066, 93055115559219BE8441880597C533381B99213B, 97C3376AB551E899F347CC9DDF49EA01DB2D7903, 99FAD0862E2E8D363F3E18952FD92E09493CC27D, A101CBCCD950AA36FC3B40C3C331FDE43ACDBBD2, A227C0A4425E24268B759A740231676A589CA4E6, A997A7AAE727D2C12CCE80FE3607317775A4DF3E, B0271CA76052EC340014D7BCCDBD69325A4E60F2, B0CD4F5DF192BFFE6500E44B80C28505DFD9CA66, B16E7D56A8DC0FF6B3AFD797E1EAB22B20DFFB39, D49979D0063B28BD73390481E6AE642C00CE0791, D518F5C648AB64B390A29AA2858219318CFC556A, DF223D653F761ED55F9C0774F1DBF545FD741F86, DF8FC5213AA11EE445EAD1AAE17A826E7D51A743, E02DD79A8CAED662969F6D5D0792F2CB283116E8, E8F4EA3857EF5FDFEC1A2063D707609251F207DB, F26CAE9E79871DF3A47FA61A755DC028C18451FC, FF09608790077E1BA52C03D9390E0805189ADAD7, A9747A3F58F8F408FECEFC48DB0A18A1CB6DACAE
Domains	api[.]tikavodot[.]co[.]il, magicallyday[.]com, processplanet[.]org
IPv4	3[.]95[.]7[.]142, 35[.]175[.]224[.]64, 51[.]16[.]209[.]105, 62[.]106[.]66[.]112, 157[.]20[.]182[.]45, 161[.]35[.]172[.]55, 167[.]99[.]224[.]13, 194[.]11[.]246[.]78, 194[.]11[.]246[.]101, 206[.]71[.]149[.]51, 212[.]232[.]22[.]136

TYPE	VALUE
Filenames	OsUpdater.exe, Blub.exe, stealer.exe, 7d1e9726b5YZPYc.dll, fe197add74IVcQn.exe, vmsvc.exe, 3a70e4c8c2IVcQn.exe, Dsync-es.exe, App_chek.exe, steam.exe, antimage.exe, wtsapi32.dll, msi.dll, WinWin.exe, 20241118_223247_Launcher.exe, Launcher.dll, vcruntime140_1.dll, ncrypt.dll, WinWin(persist).exe, 0bff183a39ruQsY.dll, 20d188afdcpfLFq.exe, dttcodexgigas.exe, 7295be2b1fHxjyf.exe, fa54125dc8ZpaNJ.exe, 20d188afdcWgOQB.exe, bd34a33f5bHOVby.exe, re.exe, 1110254b63WfTEa.exe, FMAPP.dll, bd34a33f5bJeJOf.exe, 8525e604dfKuDNr.exe, main.exe, 504f53ca8esoLmG.dll, 66f3e097e4tnyHR.exe, 7295be2b1fAzMZl.exe, ESETGO.exe, Revoke.dll, AppVs.exe



References

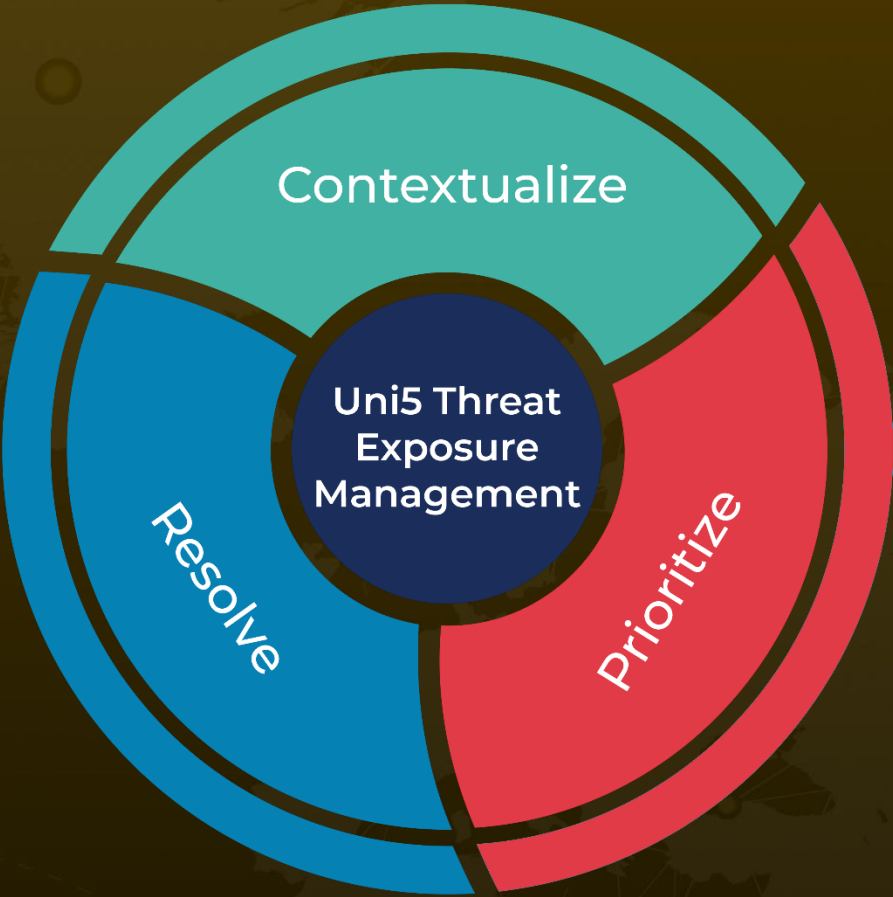
<https://www.welivesecurity.com/en/eset-research/muddywater-snakes-riverbank/>

<https://hivepro.com/threat-advisory/muddywater-deploys-phoenix-backdoor-in-targeted-espionage-campaign/>

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